S2.2 The covalent model

Covalent network structures

Learning outcomes

By the end of this section you should be able to:

- Describe allotropes in terms of structural and bonding patterns, and resulting impacts on chemical and physical properties.
- Describe the structures and explain the properties of silicon, silicon dioxide and carbon's allotropes: diamond, graphite, fullerenes and graphene.

Our understanding of carbon has evolved significantly. It began with the use of soot as a pigment thousands of years ago and progressed to the use of coal in the process of steelmaking (**Figure 1**). It is integral to radiocarbon dating, and diamonds have significance both culturally and economically. In 1996, the Nobel Prize in Chemistry was awarded to Smalley, Curl and Kroto for discovering a new form of carbon, known as C_{60} buckminsterfullerene.

Figure 1. Different Forms of Carbon.

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More information for figure 1

An interactive schematic presents four images illustrating different forms of carbon.
Image 1: Soot. A dark, powdery substance covering the surface.
Image 2: Fullerene. A hollow, closed-cage structure composed of 20 hexagons and 12 pentagons, with atoms positioned at the vertices.
Image 3: Coal. Many dark, irregularly shaped coal pieces.
Image 4: Diamond. A transparent, highly reflective piece of a diamond.
These images help users recognize and differentiate between various forms of carbon.

An entire class of compounds including various tubes and spheres have since been developed based on C_{60}. How can the same element which heats fireplaces be worth thousands in a diamond and win scientists Nobel Prizes as fullerenes?

In the big picture, S2.2.0 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-big-picture-id-43109/), we referred to two structures that are made purely of carbon: diamond and graphite. Although made of the same element, these two compounds have very different properties. It must be their bonding and the resulting structures that determine these properties.

In this section we explore this further, focusing on structures containing two elements: carbon and silicon. Both these elements can form four single bonds and have a unique ability to form bonds with themselves to create chain-like structures. This allows them to continue to form covalent bonds indefinitely, forming large lattice structures known as covalent network structures. These do not exist as discrete molecules made up of a few atoms, a 1 carat diamond would have approximately 1×10^{22} carbon atoms!

Allotropes of carbon

The ability of carbon to bond to itself repeatedly allows it to form multiple structures. Carbon can bond to itself to form diamond and graphite, as well as graphene and fullerenes. The existence of these various physical forms of the same element are known as allotropes. These have different properties such as electrical conductivity or hardness due to the type of bonding within these structures.

Diamond

Diamond is a covalent network structure made up of carbon atoms connected to each other, which theoretically can continue infinitely. Each carbon in diamond is covalently bonded to four other carbons. What molecular geometry would you expect to find in diamond? What will the bond angle be?

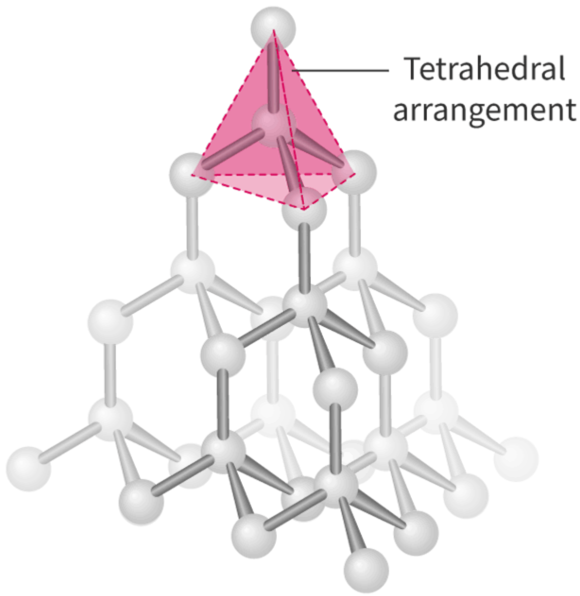

Figure 2. Molecular structure of diamond.

More information for figure 2

The image is a structural diagram illustrating the molecular structure of diamond. It shows a 3D representation where each carbon atom is covalently bonded to four other carbon atoms. The diagram highlights the tetrahedral arrangement, with its geometric shape emphasized by a pink overlay. Carbon atoms are represented as spheres connected by sticks to indicate covalent bonds, creating a repetitive pattern that continues infinitely in three dimensions. The tetrahedral geometry and bond connections form a network, demonstrating the strong and rigid structure characteristic of diamond.

[Generated by AI]

Because each carbon is surrounded by four other carbons, it would form a tetrahedral arrangement with a bond angle of 109.5º (**Figure 2**).

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These carbon–carbon covalent bonds are very strong, which means that diamond is a very hard substance, one of the hardest substances known to humankind. Do you think carbon would have a high or low melting and boiling point?

Creativity, activity, service

Strand: Service
Learning outcomes:

- Recognise and consider the ethics of choices and actions.
- Demonstrate engagement with issues of global significance.

Diamonds are precious gemstones (https://www.minerals.net/gemstone/diamond_gemstone.aspx) and throughout history have been much sought-after due to their unique properties. However, according to CBS news (https://www.cbsnews.com/news/facts-about-blood-diamonds/), between 4% and 15% of traded diamonds are conflict diamonds (https://www.globalwitness.org/en/campaigns/conflict-diamonds/), sourced from war zones and used to fund a country's war efforts.

As an IB student, what can you do at your school or in your local community to raise awareness of such global issues?

Diamond has an extremely high melting (more than 3500 °C) and boiling point (more than 4800 °C)! All of carbon's electrons are involved in forming these strong covalent bonds. This means there are no electrons free and thus diamond has poor electrical conductivity.

Graphite

Graphite contains carbon atoms which are joined to three other carbons. What do you think the molecular geometry will be for each carbon? What is the expected bond angle?

The molecular geometry of each carbon will be trigonal planar with bond angles of 120°. The carbon atoms form a continuous structure with hexagonal rings, each ring made of six carbons. This makes a planar layer. As you can see in **Figure 3**, the layers can then be stacked on top of each other.

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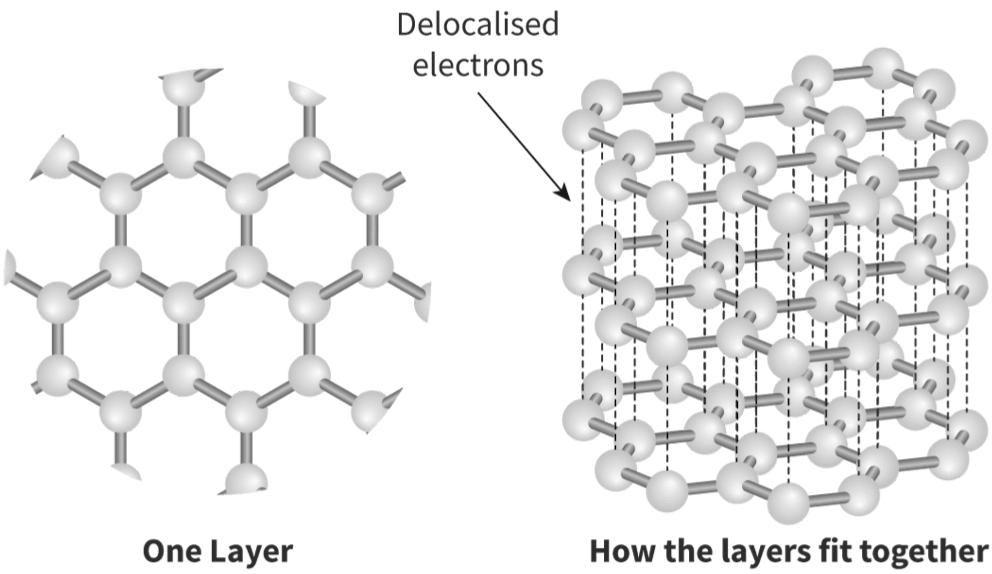

Figure 3. Molecular structure of graphite.

More information for figure 3

The image depicts the molecular structure of graphite, illustrating both a single layer and the arrangement of multiple layers. On the left, a single layer of graphite is shown with carbon atoms forming a hexagonal lattice. Each carbon atom is covalently bonded to three other carbon atoms, forming a horizontal plane. On the right side, multiple layers of these hexagonal arrangements are stacked, demonstrating how the layers fit together. There is an arrow pointing to one layer, labeled "Delocalised electrons," indicating that electrons are not bound within just one layer but can move between them. The labels "One Layer" and "How the layers fit together" provide additional context to understand the structural alignment of graphite.

[Generated by AI]

Carbon is found in group 14 of the periodic table and has four valence electrons. How many covalent bonds would you expect carbon to form to complete its octet? However, in graphite, each carbon is only bonded to three other carbons. This means one of its electrons is not involved in bonding and instead is delocalised. Each delocalised electron can be found freely moving freely between the layers in the structure.

What do you think would happen if the graphite was conducted to a battery with a positive and negative terminal? Remember that electrons are negative and will be affected by other charges around them due to the electrostatic forces of attraction.

As the electrons are free to move, they will be attracted to the positive terminal of the battery and flow along a pathway towards it, carrying their electrical charge along.

Use **Interactive 1** to see graphite show good electrical conductivity.

Interactive 1. The presence of delocalised electrons means that graphite can conduct electricity.

More information for interactive 1

This interactive consists of three slides demonstrating how a graphite pencil can conduct electricity due to the presence of delocalised electrons in its structure.
Slide 1: A simple electrical circuit is shown, which includes a battery, a light bulb, connecting wires, and a pencil connected at both ends using metal clips. The pencil acts as the conductor in the circuit. The bulb is currently off, showing that the circuit is not powered.
Slide 2: A zoomed-in illustration explains what is inside the pencil lead. It shows the graphite structure at the atomic level with multiple layers of carbon atoms arranged in hexagonal sheets. Between the layers are red dots representing delocalised electrons. These free-moving electrons are responsible for conducting electricity. One arrow flows from the pencil to the graphite layers, and another arrow points from the graphite structure to a battery, symbolising electrical conduction.
Slide 3: The same circuit from Slide 1 is shown again, but now the bulb is glowing. This indicates that the circuit is switched on and electricity is flowing through the pencil. It confirms that the graphite within the pencil is allowing electric current to pass, lighting up the bulb. This demonstrates the concept that graphite is a good conductor due to the mobility of delocalised electrons within its layers.
The interactive helps users understand that graphite conducts electricity because it contains delocalised electrons that can move freely between layers.

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The stacked layers in graphite are held together by relatively weak London dispersion forces. You will learn more about these intermolecular forces in section S2.2.9 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/strength-of-intermolecular-forces-id-43852/).

As these London dispersion forces holding layers of carbons together are fairly weak, this makes them easy to separate. This is why graphite is quite soft and slippery: the layers are easy to separate. This makes graphite ideal for use in pencils, each layer can slide off onto the page with very little force applied. This is unlike diamond. The ability of the layers in graphite structures to slide over each other also make it ideal for use as a ‘dry’ lubricant. This is particularly useful in situations where heat might be present, which might otherwise cause an oil-based lubricant to ignite.

Graphene

Graphene is composed of just one layer of graphite and is said to be the strongest, lightest and thinnest material known to humans. It is so thin, just one carbon atom thick, that it can technically be referred to as being two-dimensional. Whereas both diamond and graphite occur naturally, graphene is a synthetic carbon allotrope.

Each carbon in graphene is bonded to three other carbons with no non-bonding pairs (**Figure 4**). What would you expect the molecular geometry and bond angle to be for graphene?

> Each planar layer of carbons in graphene is made up of hexagonal rings with bond angles of 120° between the carbons, giving a trigonal planar molecular geometry. These transparent layers are incredibly strong, its tensile strength is almost 1000 times greater than that of steel.

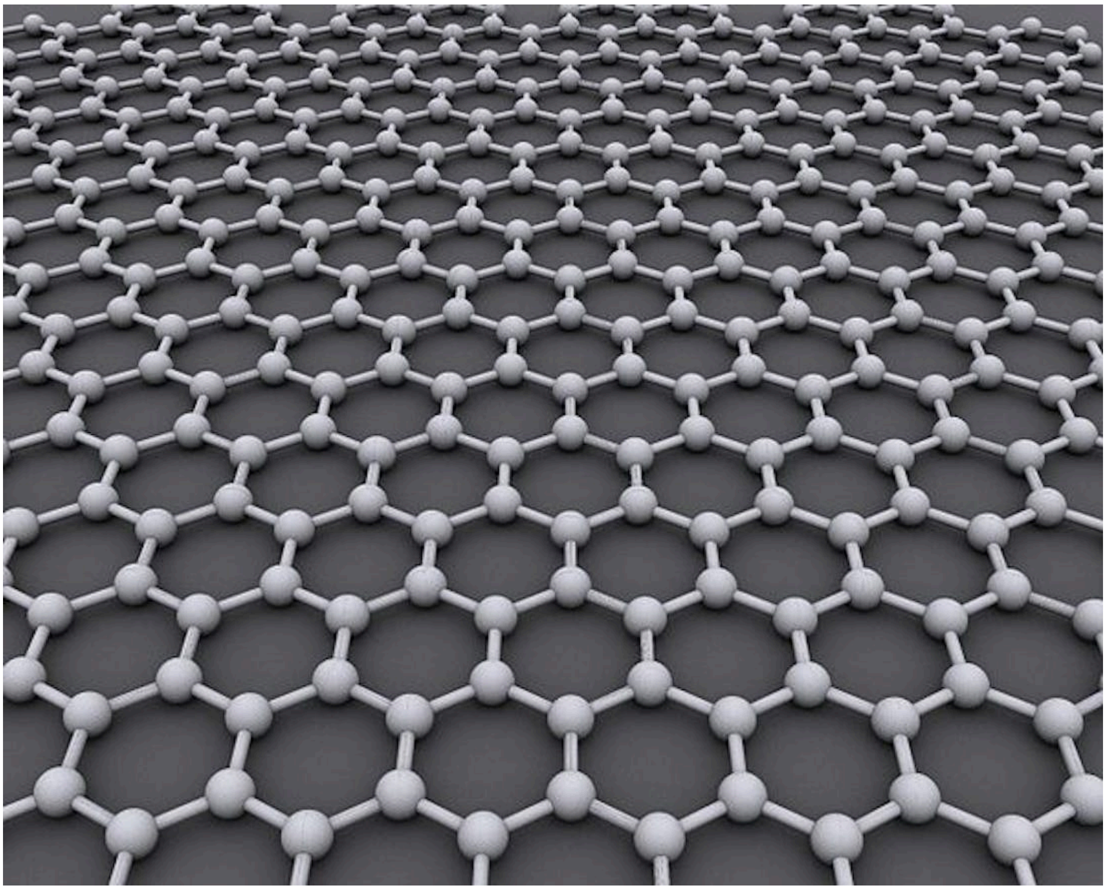

Figure 4. Molecular structure of graphene.

More information for figure 4

The image is a 3D rendering showing the molecular structure of graphene. It depicts a hexagonal lattice formed by carbon atoms. Each carbon atom is represented as a small sphere, and they are connected by straight lines indicating single bonds between them. The pattern repeats uniformly, creating a flat honeycomb structure. Graphene's structure is notable for its simplicity and symmetry, forming a two-dimensional plane that extends horizontally across the image.

[Generated by AI]

Recall that carbon is in group 14 and thus has four valence electrons. How many single bonds would you expect carbon to form to fulfil the octet rule?

However, as we identified previously, each carbon in graphene is only bonded to three other carbons. Where is the other electron found?

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The unbonded electron can move around freely and is a delocalised electron. This results in graphene having very strong electrical conductivity (**Figure 5**). In fact, it is one of the strongest conductive materials in existence.

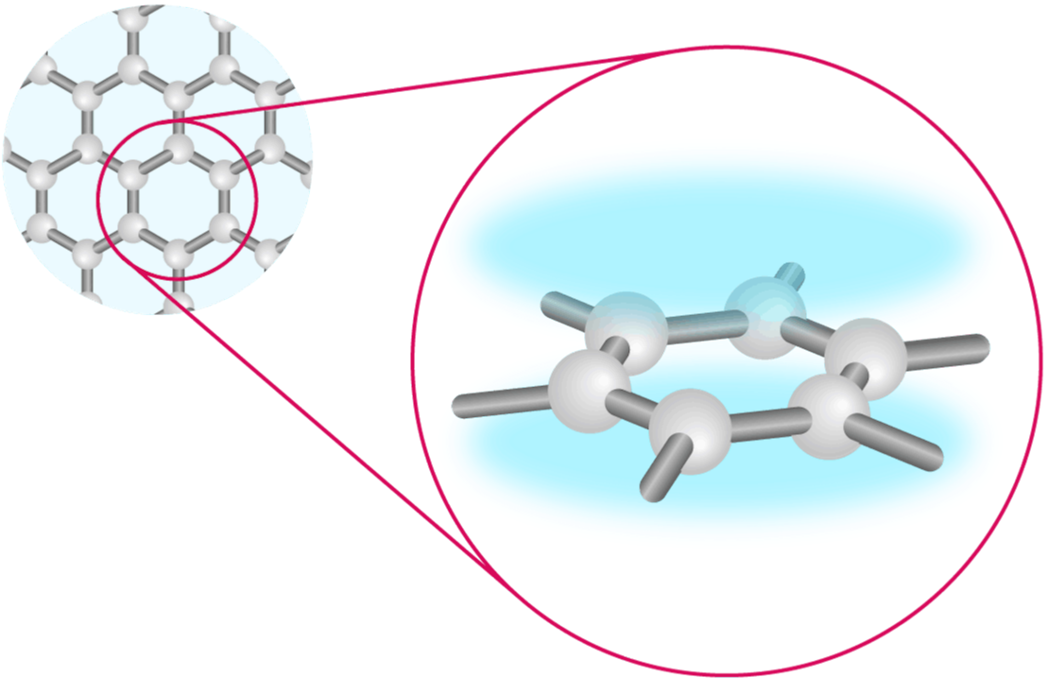


Figure 5. Graphene has a cloud of delocalised electrons found above and below the plane, which makes it highly conductive.

More information for figure 5

The image is a diagram illustrating the structure of graphene. It features a magnified view of a section of the graphene hexagonal lattice with overlapping spheres and rods representing carbon atoms and bonds, respectively. A larger blue area is depicted above and below the plane of the lattice, representing the delocalised electron cloud. This blue cloud signifies how electrons are spread across the surface, contributing to graphene's conductivity. The segment shows a direct contrast between the atomic structure and the electron distribution that affects graphene's unique properties.

[Generated by AI]

Graphene is the strongest material to ever exist. It is 100 times stronger than steel and also has very high thermal conductivity.

This property, combined with those discussed already, should give you some understanding of the huge number of potential uses for graphene in many different industries, including technology, renewable energy and product design. The discovery of these properties of graphene using groundbreaking experiments won Geim and Novoselov the Nobel Prize for Physics in 2010.

International Mindedness

Metals that are usually used in electrical appliances are extracted globally via energy intensive methods. Adding just 1% graphene to plastics could allow these plastics to conduct electricity, reducing overall emissions associated with the mining of metals. However, the benefits of this reduction in metal mining could be offset by the issues surrounding disposal of plastics. Current methods of producing graphene are expensive as well as energy and resource intensive.

Development of materials such as graphene for a particular use can have significant local and global affects.

Sheets of graphene can be rolled to create tubes, known as nanotubes (**Figure 6**). These tubes also have delocalised electrons present that make them excellent electrical conductors.

These tubes have many uses in the medical and pharmaceutical industry, as well as in electronics.

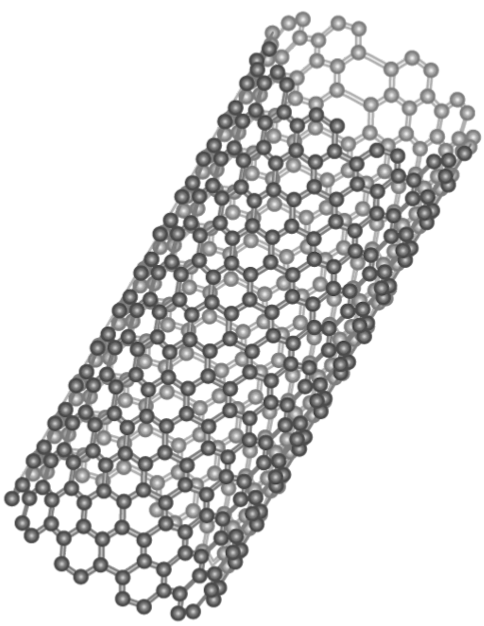


Figure 6. Molecular structure of a carbon nanotube.

More information for figure 6

The image depicts the molecular structure of a carbon nanotube. It is shown as a cylindrical arrangement of carbon atoms forming a hexagonal lattice, resembling a rolled-up sheet of graphene. The structure consists of repeating hexagonal rings, similar to honeycomb patterns, highlighting its tubular shape and intricate, repeating geometry. Each point in the lattice represents a carbon atom interconnected with others to form a stable, hollow structure. This depiction provides insight into the atomic-level configuration of carbon nanotubes, which have applications in various fields due to their strength and electrical properties.

[Generated by AI]

Fullerenes

Another allotrope of carbon are fullerenes, which are similar to graphene but the carbon atoms connect together to form spherical shapes. The first fullerene to be produced was the C_{60} buckminsterfullerene, which are often referred to 'bucky balls' as they have a similar structure to a football (soccer ball). C_{60} should give you a clue about the number of carbons involved in this molecule: 60 carbons. The fact that this is a fixed number means that although it contains quite a lot of atoms, unlike diamond and graphite, it is technically not a giant network.

Similar to graphite, each carbon in fullerene is bonded to three other carbons. This results in a trigonal planar molecular geometry and a bond angle of 120°. These form a mix of 20 hexagonal (made of 6 carbons) rings and 12 pentagonal (made of 5 carbons) rings (**Figure 7**).

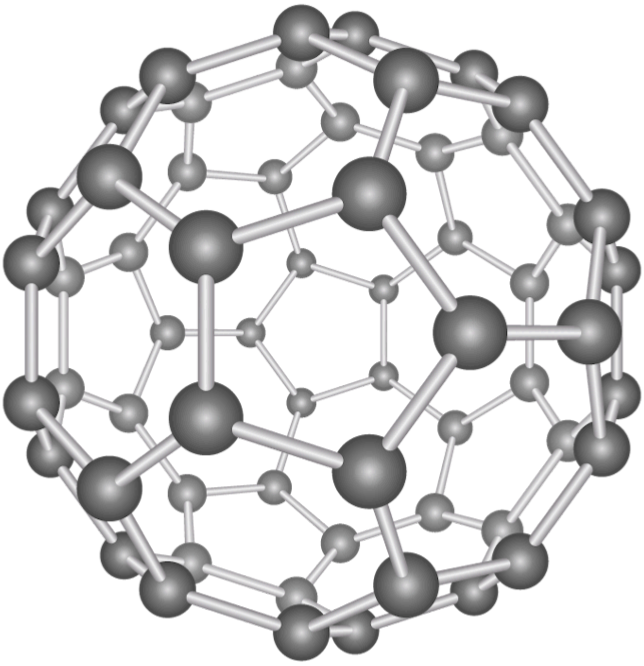


Figure 7. Molecular structure of C_{60} buckminsterfullerene.

More information for figure 7

The image shows a 3D model of a C60 buckminsterfullerene molecule, which resembles a soccer ball structure made up of connected spheres. The spheres represent carbon atoms, and the lines connecting them represent the bonds between the atoms. The structure consists of 20 hexagonal rings and 12 pentagonal rings. Each carbon atom is bonded to three others, forming a spherical shape composed entirely of carbon atoms. This geometry creates a mix of hexagons and pentagons, crucial for understanding the properties of fullerene related to its molecular geometry and electron delocalization, as discussed in the accompanying text.

[Generated by AI]

Also, like graphite, as the carbons are only bonded to three other carbons, there are delocalised electrons present in fullerene. However, its conductivity is much poorer than that of graphite. Although delocalised electrons are present, because they can only move around each fullerene and not jump between two separate fullerenes limits their conductivity because they cannot move in along a linear path as with graphite and graphene.

Both nanotubes and fullerenes have wide uses in the medical industry due to their ability to bind to specific target molecules and to carry specific drug and gene therapies to a target site (**Figure 8**).

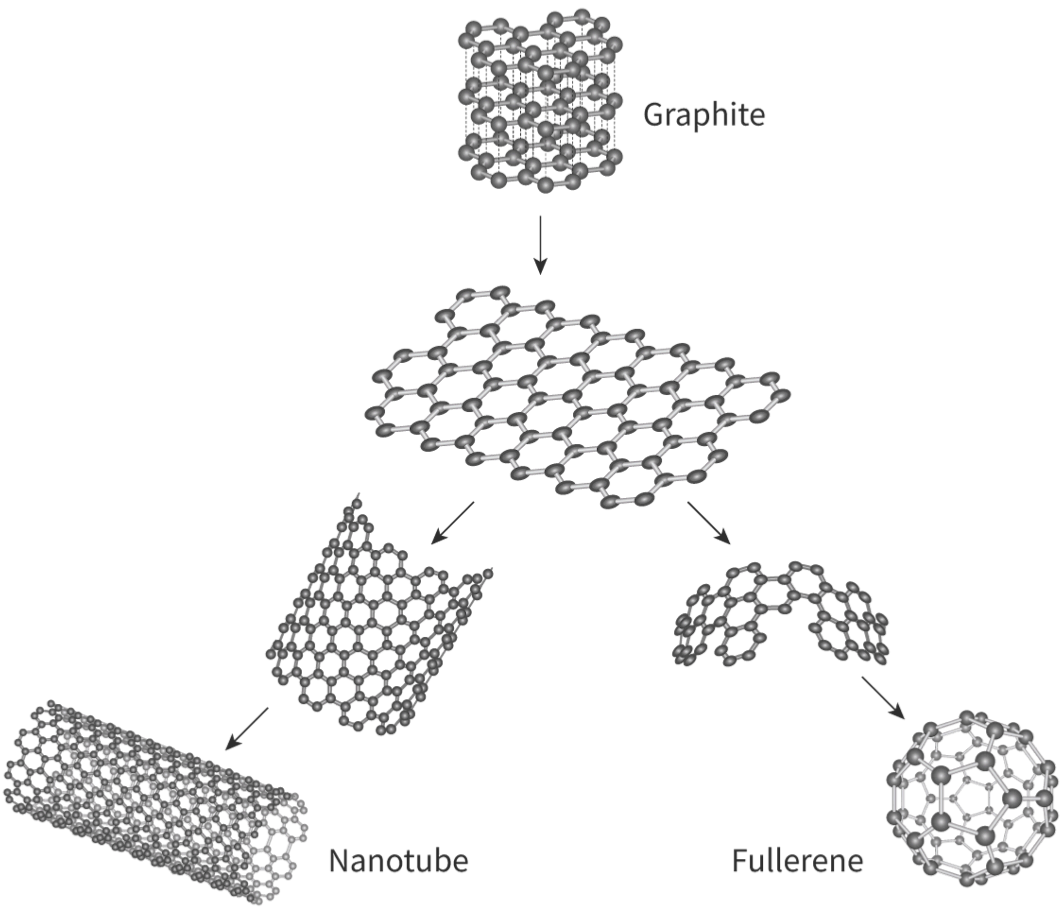


Figure 8. Trigonal planar allotropes of carbon.

More information for figure 8

The image is a diagram depicting different carbon allotropes. At the top, there is a structure labeled 'Graphite,' which features a series of stacked hexagonal rings. Below it, an arrow points to 'Graphene,' showing a single layer of hexagonal rings, which is the base structure. From this central graphene structure, arrows diverge to other forms. One arrow leads to 'Nanotube,' which shows a rolled-up sheet of graphene forming a cylindrical tube. Another arrow leads to 'Fullerene,' showing a partially folded graphene sheet and a spherical structure composed of hexagons and pentagons representing the fullerene shape.

[Generated by AI]

This next activity in **Interactive 2** will allow you to drag-and-drop molecular geometries, bond angles, properties and uses of each of the carbon allotropes described above.

	Diamond	Graphite	Graphene	Fullerene C_{60}	Nanotubes
Image					
Molecular geometry					
Bond angle					
Electrical conductivity					
Special properties					
Uses					

Drug delivery (medical and pharmaceuticals)

Technology; renewable energy (photovoltaic cells)

Jewellery; tools and machinery for cutting glass

Medical and pharmaceuticals; electronics

Dry lubricant; pencils

120°

109.5°

Hardest natural substance known to man

High thermal conductivity, very flexible with high tensile strength

Just one atom thick – the only 2D material to exist 100 times stronger than steel

Very stable and can withstand high temperatures and high pressures

Soft and slippery as layers can slide over each other

Sphere of trigonal planar

Layers of trigonal planar

One layer of trigonal planar

Tube (roll) of trigonal planar

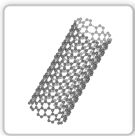
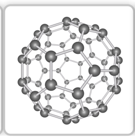
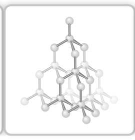
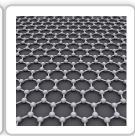
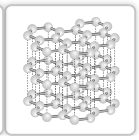
Tetrahedral

Very good electrical conductor

Good electrical conductor

Semiconductor

Not an electrical conductor (insulator)

Check

Interactivity 2. Features of Carbon Allotropes.

More information

An interactive table helps learners explore and compare the structural and functional characteristics of various carbon allotropes. Users must drag and drop the correct options into the missing fields of the table, including the image, molecular geometry, bond angle, electrical conductivity, special properties, and uses for each of the following carbon allotropes: diamond, graphite, graphene, fullerene C_{60}, and carbon nanotubes.
The drag-and-drop options are as follows:
Drug delivery (medical and pharmaceuticals)
Technology; renewable energy (photovoltaic cells)
Jewellery; tools and machinery for cutting glass
Medical and pharmaceuticals; electronics
Dry lubricant; pencils
$120°$
$109.5°$
Hardest natural substance known to man
High thermal conductivity, very flexible with high tensile strength
Just one atom thick - the only 2D material to exist 100 times stronger than steel
Very stable and can withstand high temperatures and high pressures
Soft and slippery as layers can slide over each other
Sphere of trigonal planar
Layers of trigonal planar
One layer of trigonal planar
Tube (roll) of trigonal planar
Tetrahedral
Very good electrical conductor
Good electrical conductor
Semiconductor
Not an electrical conductor (insulator)
A three-dimensional hollow cylindrical tube with walls made of hexagonal carbon rings in a honeycomb pattern
A three-dimensional hollow, cage-like, spherical structure composed of 12 pentagons and 20 hexagons
A three-dimensional lattice of carbon atoms arranged in a tetrahedral geometry. Each carbon atom is bonded to four other carbon atoms
A single layer of carbon atoms arranged in a hexagonal honeycomb pattern
A three-dimensional layered structure composed of stacked hexagonal layers of carbon atoms, with layers connected by dashed lines.
This interactive helps users identify and compare the structural features, physical properties, electrical conductivity, and practical applications of different carbon allotropes.
Read below for the solution:
Diamond:
Image: A three-dimensional lattice of carbon atoms arranged in a tetrahedral geometry. Each carbon atom is bonded to four other carbon atoms
Molecular geometry: Tetrahedral
Bond angle: $109.5°$
Electrical conductivity: Not an electrical conductor (insulator)
Special properties: Hardest natural substance known to man
Uses: Jewellery; tools and machinery for cutting glass
Graphite:

Image: A three-dimensional layered structure composed of stacked hexagonal layers of carbon atoms, with layers connected by dashed lines.
Molecular geometry: Layers of trigonal planar
Bond angle: $120°$
Electrical conductivity: Good electrical conductor
Special properties: Soft and slippery as layers can slide over each other
Uses: Dry lubricant; pencils
Graphene:
Image: A single layer of carbon atoms arranged in a hexagonal honeycomb pattern
Molecular geometry: One layer of trigonal planar
Bond angle: $120°$
Electrical conductivity: Very good electrical conductor
Special properties: Just one atom thick - the only 2D material to exist 100 times stronger than steel
Uses: Technology; renewable energy (photovoltaic cells)
Fullerene C_{60}:
Image: A three-dimensional hollow, cage-like, spherical structure composed of 12 pentagons and 20 hexagons
Molecular geometry: Sphere of trigonal planar
Bond angle: $120°$
Electrical conductivity: Semiconductor
Special properties: Very stable and can withstand high temperatures and high pressures
Uses: Drug delivery (medical and pharmaceuticals)
Nanotubes:
Image: A three-dimensional hollow cylindrical tube with walls made of hexagonal carbon rings in a honeycomb pattern
Molecular geometry: Tube (roll) of trigonal planar
Bond angle: $120°$
Electrical conductivity: Very good electrical conductor
Special properties: High thermal conductivity, very flexible with high tensile strength
Uses: Medical and pharmaceuticals; electronics.

Covalent network structures of silicon

Silicon is the second most abundant element in the Earth's crust, after oxygen. It can be found as pure silicon and also as silicon dioxide, SiO_2, commonly known as silica or quartz. Pure silicon is often used as a semiconductor in computer chips and silicon dioxide has many uses in everyday life. Glass and sand are two of the most commonly used forms of silicon dioxide.

Silicon

Pure silicon contains one silicon atom that is covalently bonded to four other silicon atoms. Analyse **Figure 9**. Where have you seen a similar structure previously?

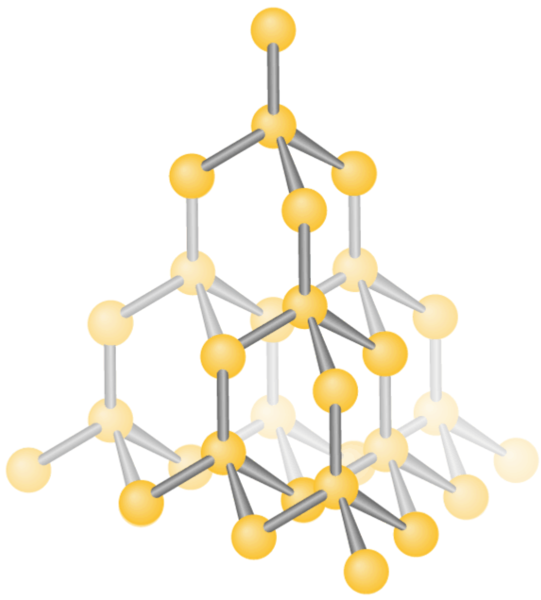


Figure 9. The covalent network structure of silicon.

More information for figure 9

This diagram depicts a three-dimensional covalent network structure of silicon atoms. Each silicon atom is illustrated as a yellow sphere, interconnected by gray rods that signify covalent bonds. Overall, the atoms are arranged in a tetrahedral pattern, where each silicon atom is covalently bonded to four others, forming a repetitive structure typical of silicon networks. The visualization emphasizes the symmetry and regularity in the arrangement, reflecting the crystalline lattice of pure silicon.

[Generated by AI]

This is like diamond. Pure silicon has a covalent network structure with each central silicon atom having a tetrahedral molecular geometry and a bond angle of 109.5°.

Why do you think carbon and silicon have an ability to form these similar structures?

Similar to carbon, silicon is also in group 14 and has four valence electrons. This means it can also form 4 covalent bonds with itself, or with other elements to form a covalent network structure.

All of silicon's valence electrons are involved in bonding so its electrical conductivity is poor. These bonds formed are relatively strong, giving silicon quite a high melting and boiling point.

Making connections

The melting point of silicon is 1410 °C whereas diamond and graphite have melting points of more than 3500 °C. Try to explain this difference in bond strength using your knowledge of bond strength from section S2.2.3 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/coordination-bonds-id-43594/).

International Mindedness

Silicon Valley is a region in both Bangalore in India and in San Francisco, California in the USA. They have been named 'Silicon Valley' due to the high concentration of technology-based companies that extensively use electronic chips, many of which are made of silicon. Although silicon lacks the delocalised electrons usually used to conduct electricity, a process known as 'doping' (introducing certain impurities to the structure) causes it to become a viable conductor, hence the name semiconductor. The silicon used in these chips is mostly produced in China. This international cooperation is essential in continuing to develop the technologies we rely on today.

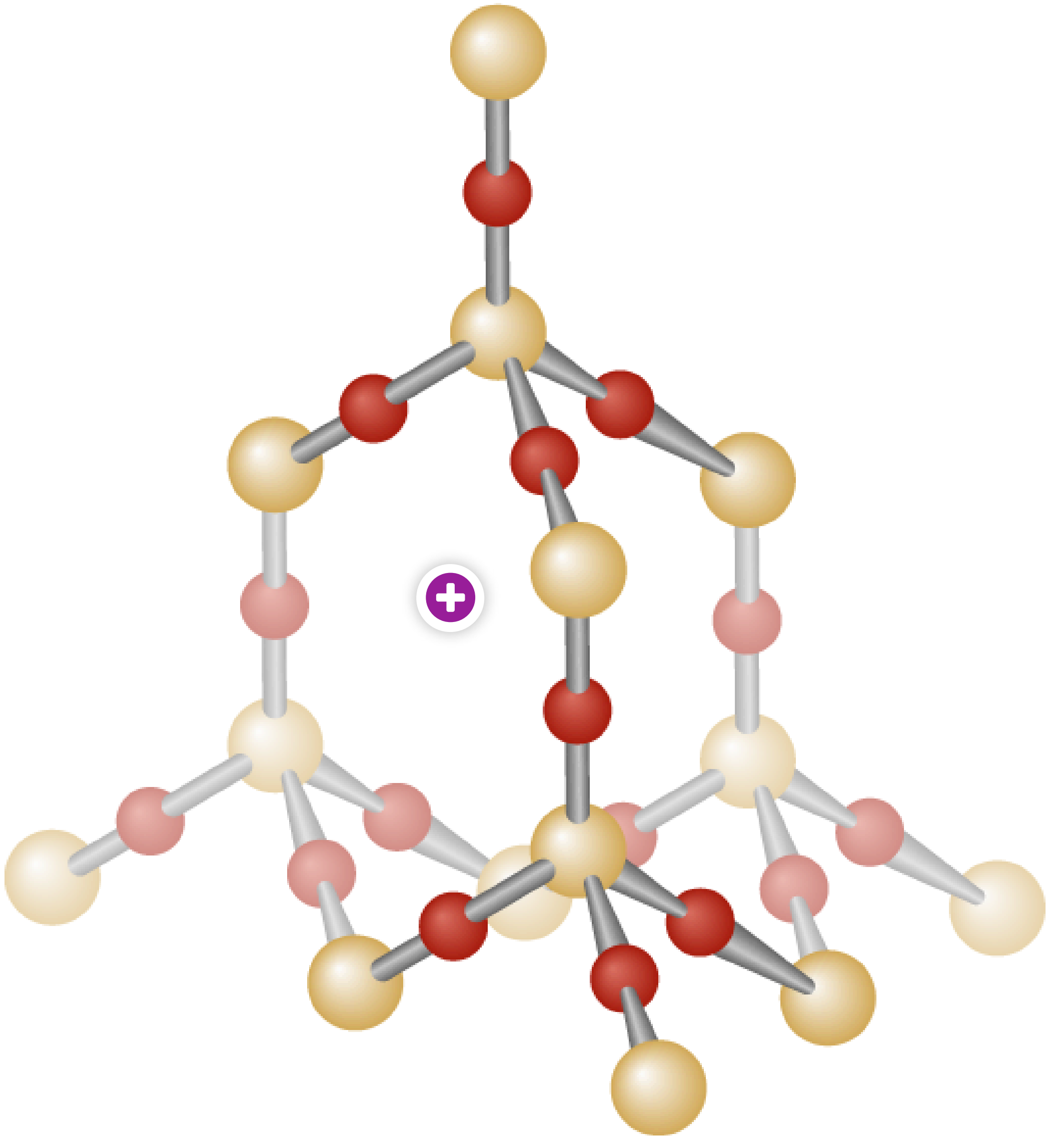
Silicon dioxide

Silicon can also form a covalent network structure with oxygen in silicon dioxide.

Silicon dioxide, SiO_2, is composed of a silicon bonded to four oxygen atoms with each oxygen also bonded to 2 silicon atoms. SiO_2 is the simplest ratio, or empirical formula, used to represent this.

As each silicon atom bonds with four oxygen atoms, it forms a tetrahedral geometry with bond angles of 109.5° (**Figure 9**).

Similar to the other substances with covalent network structures, the covalent bonds here are strong and significant amounts of energy are required to break them; hence the melting and boiling of silicon dioxide are very high.



Interactive 4. Covalent Network Structure of Silicon Dioxide.

More information for interactive 4

An interactive illustration displays a three-dimensional ball-and-stick model of silicon dioxide in two layers. Silicon is represented by large spheres, while oxygen is represented by small spheres. The model shows that each silicon atom is bonded to four oxygen atoms in a tetrahedral geometry, and each oxygen atom is bonded to two silicon atoms.
A plus sign indicates an interactive hotspot. Selecting this hotspot reveals a three-dimensional ball-and-stick model of silicon dioxide with four layers. Next to this is a

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close-up view of the arrangement of the silicon and oxygen atoms.
The bonding between the silicon and oxygen atoms is covalent.
This illustration helps users understand the structure and bonding in SiO_2.

In this next activity, you will create three-dimensional models of the covalent network structures; diamond and graphene. Creating these three-dimensional models should help you better visualise these different macromolecules and remember the differences between them.

Activity

- **IB learner profile attribute:** Knowledgeable
- **Approaches to learning:** Thinking skills — Designing procedures and models
- **Time required to complete activity:** 20 minutes
- **Activity type:** Individual and group activity

Materials:

- Toothpicks or straws
- Styrofoam balls, cotton balls or plasticine
- Protractor

Instructions:

- Using the Styrofoam balls (or cotton balls/plasticine) as atoms and toothpicks (or straws) as bonds, create your own 3D models of both diamond and graphene.
- Pair up with 3—4 other people. Using your graphenes as layers, make graphite.
- In your group, work together to create a buckminsterfullerene.
- Now, work together to use a protractor to measure the angles between the carbons. Check that the models you have made are true to what you learned about the shapes and bond angles that VSEPR theory can be used to predict.